

AWS LAB 1 - 3 Introduction to IAM, VPC, and EC2

By Rui Yang



Purpose:

Lab 1: To explore the limits of the **IAM** and experiment with some aspects of the **IAM** such as configuring user access, groups, and routing table. We also confirm that this is applied through logging in on guest accounts with set access in order to confirm whether the configurations were applied or not.

Lab 2: The purpose of this lab is to explore different aspects of a Virtual Private Cloud (VPC) and produce a custom designed network that fits to one's needs. Then we would configure and run a web server through an EC2 instance to run a subnet set within the VPC.

Lab 3: A basic overview of launching, changing storage size of the instance, managing an **EC2** with additional settings such as terminal protection and stop protection. Then we will launch the instance and creating security groups, and attempt at monitoring the EC2 through logs and get instance screenshots.

Background Information:

Amazon Web Services (AWS) is a secure cloud platform that offers a broad set of global cloud-based products. Because these products are delivered over the internet, you have on-demand access to the computer, storage, network, database, and other IT resources that you might need for your projects—and the tools to manage them. You can immediately provide and launch AWS resources. The resources are ready for you to use in minutes.

AWS Identity and Access Management (IAM) is a web service that enables Amazon Web Services (AWS) customers to manage users and user permissions in AWS. With IAM, you can centrally manage **users**, **security credentials** such as access keys, and **permissions** that control which AWS resources users can access. It is a tool that centrally manages access to launching, configuring, managing, and terminating resources in your AWS account. It also provides granular control over access to resources, including the ability to specify exactly which API calls the user is authorized to make to each service. Whether you use the AWS Management Console, the AWS CLI, or the AWS software development kits (SDKs), every call to an AWS service is an API call. With IAM, you can manage which resources can be accessed by who, and how these resources can be accessed, such as granting different

permissions to different people for different resources. For example, you might allow some users full access to Amazon EC2, Amazon S3, Amazon DynamoDB, Amazon Redshift, and other AWS services, but not allow other users to access those services.

Amazon Virtual Private Cloud (Amazon VPC) enables you to launch Amazon Web Services (AWS) resources into a pre-defined virtual network. This virtual network is very similar to a traditional physical network that would be operated in one's own data center, with additional benefits of using the scalable infrastructure of AWS. You can create a VPC that spans multiple Availability Zones.

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides resizable compute capacity in the cloud. It is designed to make web-scale cloud computing easier for developers.

Amazon EC2's simple web service interface allows for developers to obtain and configure capacity with minimal friction. It provides developers with control of your computing resources and runs on Amazon's proven computing environment. Amazon EC2 reduces the time required to obtain and boot new server instances to just a few minutes, allowing developers to quickly scale capacity, both increase and decrease, as needs change.

Amazon EC2 changes the economics of computing as it only requires developers to pay for what they use, instead of the entire set including unused capacity. Amazon EC2 provides developers the tools to build failure resilient applications and isolate themselves from common failure scenarios.

LAB 1 PROCESS -

Users before having access to a group and therefore has no permissions -

Remove

Add user to groups

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.

☐	Group name	▲	Attached polici...	▼
No resources				
This user does not belong to any groups.				

Adding User 1 to S3-Support:

Add user user-1 to groups

Select which groups to add the user user-1.

Other groups (1/3)

Create group

☒

Group name

▲

Users

▼

Attached polici...

▼

Created

▼

☐

EC2-Admin

0

-

2025-01-...

☐

EC2-Support

0

AmazonEC2ReadO...

2025-01-...

☒

S3-Support

0

AmazonS3ReadOn...

2025-01-...

Cancel

Add user to group(s)

Confirms that user 1 is now part of S3-Support group -

Remove

Add user to groups

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.

☐	Group name	▲	Attached polici...	▼
☐	S3-Support		AmazonS3ReadO...	

Adding User 2 into S2-Support Group -

Add user user-2 to groups

Select which groups to add the user user-2.

Other groups (1/3)

Search

< 1 >

☐	Group name	Users	Attached polici...	Created
☐	EC2-Admin	0	-	2025-01-...
☒	EC2-Support	0	AmazonEC2ReadO...	2025-01-...
☐	S3-Support	0	AmazonS3ReadOn...	2025-01-...

Cancel

Add user to group(s)

Confirm User 2 is added into S2-Support group -

Permissions

Groups

(1)

Tags

Security cre

User groups membership

Remove

Add user to groups

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.

☐	Group name	Attached polici...
☐	EC2-Support	AmazonEC2Read...

Add User 3 to EC2-Admin group -

Add user user-3 to groups

Select which groups to add the user user-3.

Other groups (1/3)

Search

< 1 >

☐	Group name	Users	Attached polici...	Created
☒	EC2-Admin	0	-	2025-01-...
☐	EC2-Support	0	AmazonEC2ReadO...	2025-01-...
☐	S3-Support	0	AmazonS3ReadOn...	2025-01-...

Cancel

Add user to group(s)

Confirmation -

< | Permissions | **Groups** (1) | Tags (1) | Security cre | >

User groups membership

[Remove](#) [Add user to groups](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.

☐	Group name	▲	Attached polici...	▼
☐	EC2-Admin		-	

Additional confirmation that users were added to groups -

User groups (3) [Info](#) [Delete](#) [Create group](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Q Search

< 1 > ⚙

☐	Group name	▲	Users	▼	Permissions	▼
☐	EC2-Admin		1		Defined	
☐	EC2-Support		1		Defined	
☐	S3-Support		1		Defined	

Signing into user-1 from an incognito tab

Amazon Web Services Sign-In

https://us-...

IAM user sign in

Account ID (12 digits) or account alias

304388171606

IAM username

user-1

Password

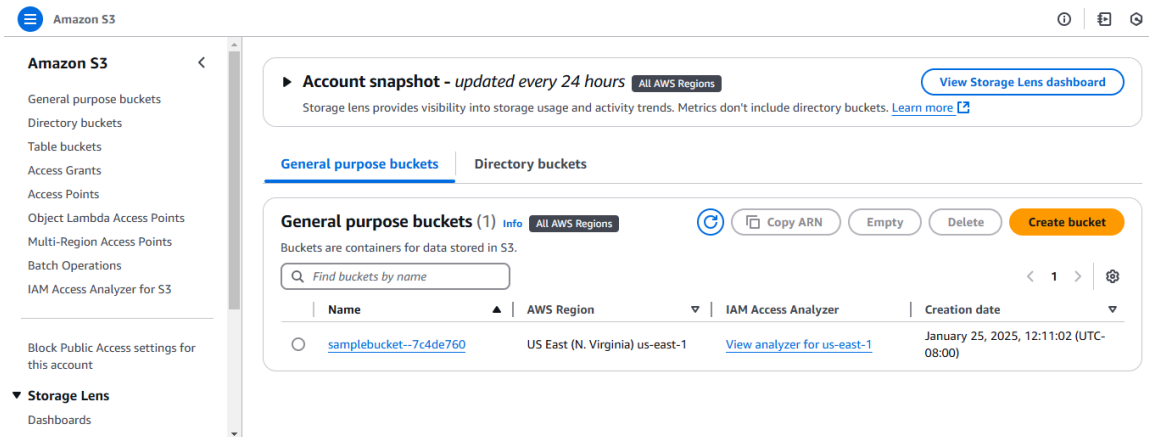
☐ Show Password [Having trouble?](#)

[Sign in](#)

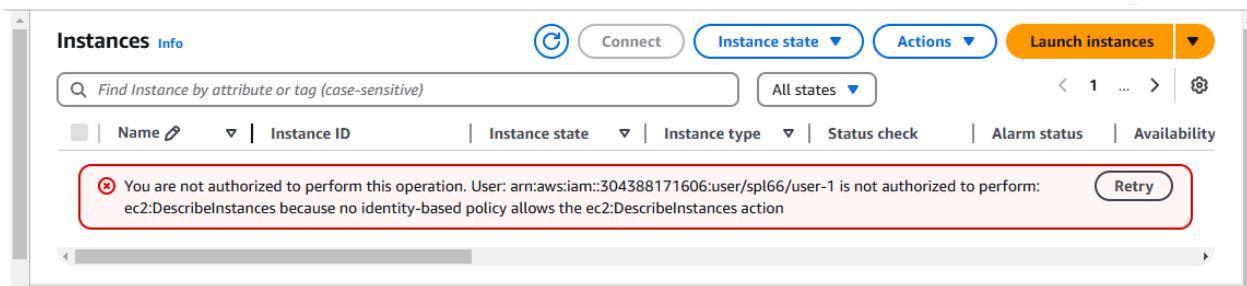
[Sign in using root user email](#)

[Create a new AWS account](#)

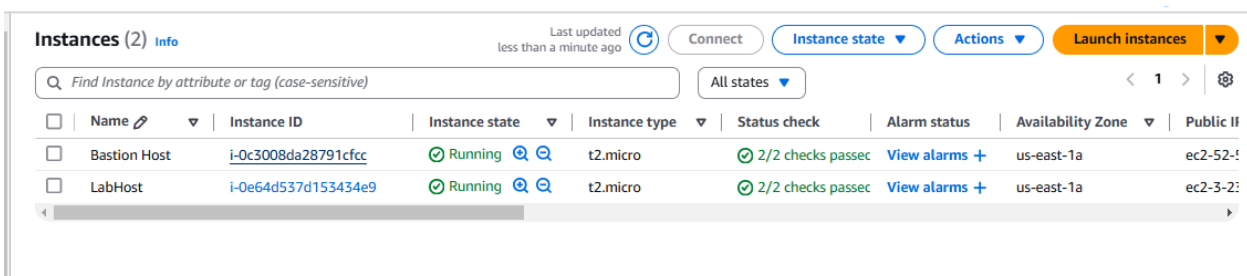
User 3 can access S3 buckets –



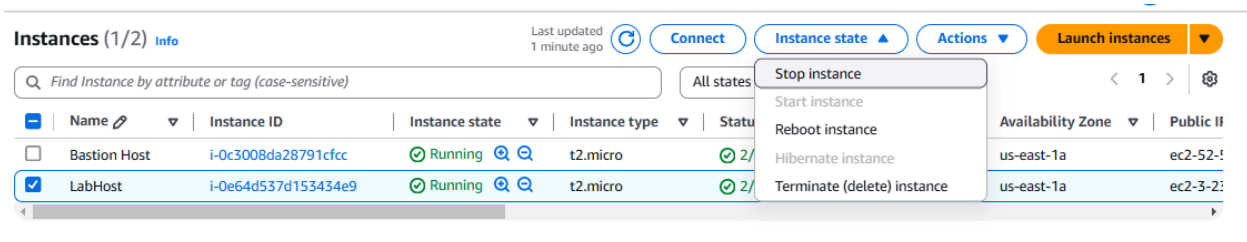
But they cannot access EC2 Instances –



After signing into User 2, we can see that user 2 is able to access EC2 Instances –



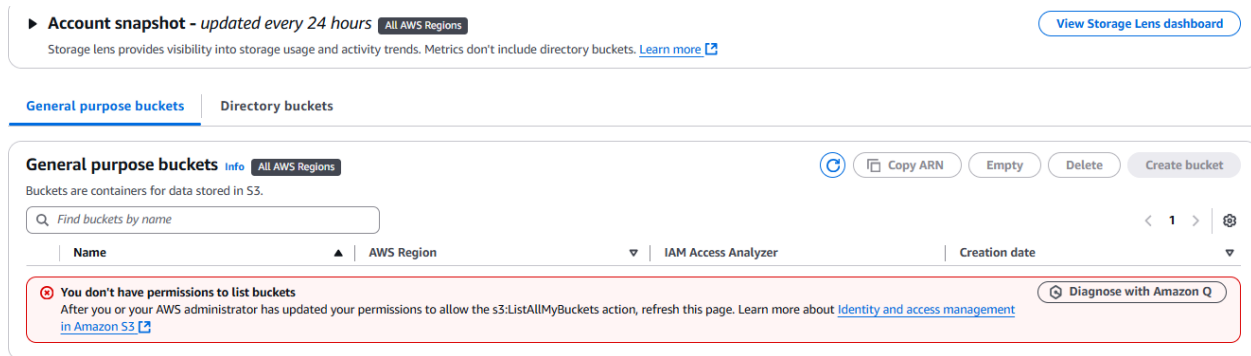
User 2 attempting to stop the instance “LabHost” –



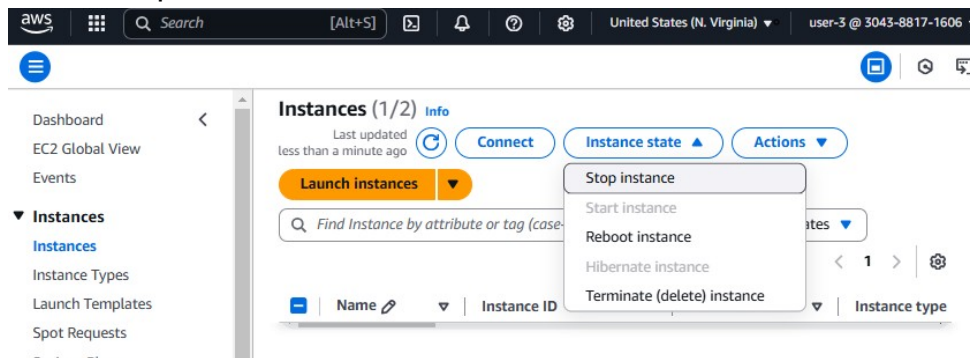
But is rejected due to being authorized with only view permissions and not ability to start of stop instances -



As user 2 only have access to view EC2, they are therefore not allowed to view S3 –



Now we log into User 1, which has the ability to start and stop EC2 as admin, tries to stop the instance “LabHost” -



User 1 is able to stop “Labhost” – It enters the “stopping” state and then enters the “stopped” state –

The screenshot shows the AWS Management Console 'Instances' page. It displays two EC2 instances: 'Bastion Host' (ID: i-0c3008da28791cfcc, state: Running) and 'LabHost' (ID: i-0e64d537d153434e9, state: Stopped). The 'LabHost' instance is selected. Below the table, there are buttons for 'Connect', 'Instance state', 'Actions', and 'Launch instances'. A search bar and a table of instance details are also visible.

Name	Instance ID	Instance state	Instance type
Bastion Host	i-0c3008da28791cfcc	Running	t2.micro
LabHost	i-0e64d537d153434e9	Stopped	t2.micro

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
Bastion Host	i-0c3008da28791cfcc	Running	t2.micro	2/2 checks passed	User: arn:aws:iam::us-east-1a	us-east-1a	ec2-52-1
LabHost	i-0e64d537d153434e9	Stopped	t2.micro	2/2 checks passed	User: arn:aws:iam::us-east-1a	us-east-1a	-

Lab Summary –

Experimented with adding users into groups that give them different accessibility and henceforth limit their access to only Amazon AWS Services they are supposed to use and confirmed it by logging into guest accounts and checking that it was applied.

Problems –

There was an issue where it said my login was expired or something similar and wasn't allowing me to access the console, and I had to restart the lab.

Conclusion –

This lab introduced us to the basic AWS IAM's features and abilities, including user control of Amazon Services and being able to control their access and abilities on certain specific services.

Completion–

The screenshot shows the AWS IAM console 'Lab 1: Introduction to AWS IAM' completion screen. It displays a list of tasks and their completion status. The total score is 40/40.

Task	Score
Total score	40/40
TASK 2a - Added user-1 to S3-Support group	5/5
TASK 2b - Added user-2 to EC2-Support group	5/5
TASK 2c - Added user-3 to EC2-Admin group	5/5
TASK 3a - user-1 logged in	5/5
TASK 3b - user-2 logged in	5/5
TASK 3c - user-2 ec2 stop instance attempt	5/5
TASK 3d - user-3 logged in	5/5
TASK 3e - user-3 EC2 stop instance attempt	5/5

LAB 2 Process -

Creating a VPC -

[VPC](#) > [Your VPCs](#) > Create VPC

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only

☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

lab

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0/1665,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)

Default

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

123

► Customize AZs

Number of public subnets [Info](#)
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

01

Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

012

▼ Customize subnets CIDR blocks

Public subnet CIDR block in us-east-1a

10.0.0/24256 IPs

Private subnet CIDR block in us-east-1a

10.0.1.0/24256 IPs

NAT gateways (\$) [Info](#)
Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway

NoneIn 1 AZ1 per AZ

VPC endpoints [Info](#)
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

NoneS3 Gateway

DNS options [Info](#)

☒ Enable DNS hostnames

☒ Enable DNS resolution

► Additional tags

Preview

VPC [Show details](#)

Your AWS virtual network

lab-vpc

Cancel

Preview code

Create VPC

Creating Additional Subnets -

- Public -

The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Subnets > Create subnet'. A dropdown menu shows 'Create subnets in this VPC:' with the selected value 'vpc-0245097718e571275 (lab-vpc)'. Below this, the 'Associated VPC CIDRs' section shows 'IPv4 CIDRs' as '10.0.0.0/16'. The 'Subnet settings' section is titled 'Subnet 1 of 1'. It includes a 'Subnet name' field with the value 'lab-subnet-public2'. The 'Availability Zone' is set to 'US East (N. Virginia) / us-east-1b'. The 'IPv4 VPC CIDR block' is '10.0.0.0/16'. The 'IPv4 subnet CIDR block' is '10.0.2.0/24', with a note indicating '256 IPs'.

- Private -

The screenshot shows the 'Create subnet' page in the AWS Management Console for a private subnet. The breadcrumb navigation is 'VPC > Subnets > Create subnet'. A dropdown menu shows 'Create subnets in this VPC:' with the selected value 'vpc-0245097718e571275 (lab-vpc)'. Below this, the 'Associated VPC CIDRs' section shows 'IPv4 CIDRs' as '10.0.0.0/16'. The 'Subnet settings' section is titled 'Subnet 1 of 1'. It includes a 'Subnet name' field with the value 'lab-subnet-private2'. The 'Availability Zone' is set to 'US East (N. Virginia) / us-east-1b'. The 'IPv4 VPC CIDR block' is '10.0.0.0/16'. The 'IPv4 subnet CIDR block' is '10.0.2.0/24', with a note indicating '256 IPs'. The footer of the page includes 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Associating the newly created subnets with routing tables to make sure traffic goes to the correct location-

- Private subnet -

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	lab-subnet-private1-us-...	subnet-05d5c48476886...	10.0.1.0/24	-	rtb-069c20a6729c2c7a6 / lab-
☐	lab-subnet-public1-us-e...	subnet-0953178ad35bc...	10.0.0.0/24	-	rtb-00fef810dcf6ef44 / lab-rt
☒	lab-subnet-private2	subnet-073bfcfaa581c3...	10.0.3.0/24	-	Main (rtb-062e14fe392ef0bb1)
☐	lab-subnet-public2	subnet-05a47854cfa32...	10.0.2.0/24	-	Main (rtb-062e14fe392ef0bb1)

Selected subnets

subnet-05d5c48476886ab9d / lab-subnet-private1-us-east-1a X subnet-073bfcfaa581c341a / lab-subnet-private2 X

Cancel Save associations

- Public subnets -

VPC > Route tables > rtb-00fef810dcf6ef44 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	lab-subnet-private1-us-...	subnet-05d5c48476886...	10.0.1.0/24	-	rtb-069c20a6729c2c7a6 / lab-
☒	lab-subnet-public1-us-e...	subnet-0953178ad35bc...	10.0.0.0/24	-	rtb-00fef810dcf6ef44 / lab-rt
☐	lab-subnet-private2	subnet-073bfcfaa581c3...	10.0.3.0/24	-	rtb-069c20a6729c2c7a6 / lab-
☒	lab-subnet-public2	subnet-05a47854cfa32...	10.0.2.0/24	-	Main (rtb-062e14fe392ef0bb1)

Selected subnets

subnet-0953178ad35bcdef2 / lab-subnet-public1-us-east-1a X subnet-05a47854cfa32376a / lab-subnet-public2 X

Cancel Save associations

Route table showing that subnets were added -

Route tables (1/6) Info

Last updated less than a minute ago

Actions

Create route table

Find resources by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associa
☐	-	rtb-062e14fe392ef0bb1	-	-
☐	-	rtb-0dea33926817002ae	-	-
☐	-	rtb-08fe18d3cee9c94cd	-	-
☐	Work Public Route Table	rtb-029ae1dcd5ff31bfd	subnet-04328dbf5c59c3...	-
☐	lab-rtb-private1-us-east-1a	rtb-069c20a6729c2c7a6	2 subnets	-
☒	lab-rtb-public	rtb-00fef810dcf6ef44	2 subnets	-

Creating a Security Group/ Virtual Firewall -

VPC > Security Groups > Create security group

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

Name cannot be edited after creation.

Description Info

VPC Info

vpc-0245097718e571275 (lab-vpc) ▼

Inbound rules Info

Inbound rule 1 Delete

Type Info	Protocol Info	Port range Info
HTTP ▼	TCP	80
Source type Info	Source Info	Description - optional Info
Anywhere-IPv4 ▼	0.0.0.0/0 X	Permit web requests

Confirmation -

✓ Security group (sg-06b19c4314ce734d4 | Web Security Group) was created successfully ✕

[▶ Details](#)

sg-06b19c4314ce734d4 - Web Security Group Actions ▼

Details

Security group name Web Security Group	Security group ID sg-06b19c4314ce734d4	Description Enable HTTP access	VPC ID vpc-0245097718e571275
Owner 377965143576	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Sharing - new](#) | [VPC associations - new](#) | [Tags](#)

Inbound rules (1)

Manage tags Edit inbound rules

< 1 >

⚙

☐	Name	Security group rule ID	IP version	Type
☐	-	sgr-05d6ad0dcea0a111f	IPv4	HTTP

Launching a Web Server Instance -

Creating an instance -

EC2 > Instances > Launch an instance

Name and tags [Info](#)

Name

Web Server 1

Add additional tags

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Q Search our full catalog including 1000s of application and OS images

RecentsQuick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI

ami-0ac4dfaf1c5c0cce9 (64-bit (x86), uefi-preferred) / ami-00d9a6d7d54864374 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

Create new key pair

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0245097718e571275 (lab-vpc)
10.0.0.0/16



Subnet [Info](#)

subnet-05a47854cfa32376a
VPC: vpc-0245097718e571275 Owner: 377965143576 Availability Zone: us-east-1b
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.2.0/24

lab-subnet-public2



[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable



Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups



[Compare security group rules](#)

Web Security Group sg-06b19c4314ce734d4
VPC: vpc-0245097718e571275



Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Configure storage [Info](#)

[Advanced](#)

1x GiB Root volume 3000 IOPS (Not encrypted)

[i](#) Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage



[Add new volume](#)

[Click refresh to view backup information](#)

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.



0 x File systems

[Edit](#)

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
# Install Apache Web Server and PHP
dnf install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-
vpc/s3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```

☐ User data has already been base64 encoded

Confirmation -

A green success banner at the top states: "Success Successfully initiated launch of instance (i-035526ab64118ba88)". Below it is a "Launch log" link.

The "Instances (2)" section shows a table of running instances. The table has columns for Name, Instance ID, Instance state, Instance type, and Status check. Two instances are listed: "Bastion Host" and "Web Server 1", both in a "Running" state with "2/2 checks passed".

Name	Instance ID	Instance state	Instance type	Status check
Bastion Host	i-0dff56b570621c70d	Running	t2.micro	2/2 checks passed
Web Server 1	i-035526ab64118ba88	Running	t2.micro	2/2 checks passed

Meta-data Values -

The browser shows a "Not secure" warning and the URL "http://ec2-54-80-113-56.compute-1.amazonaws.com". The page content includes a table of meta-data values and a CPU load indicator.

Meta-Data	Value
Instanceid	i-035526ab64118ba88
Availability Zone	us-east-1b

Current CPU Load: 5%

Lab Summary -

Experimented and Created a Virtual Private Cloud (VPC) and its different features such as creating additional subnets and associating those new

subnets within the routing table and creating a virtual firewall (also known as security group) with custom configurations. Also learned how to launch a web server based off an EC2 instance.

Issues -

There weren't really issues present, the lab was finished the first time through and everything was smooth sailing. Although the same issue of logging in occurred at the start of the lab.

Conclusion -

This lab introduced us to the Amazon VPC and a very basic introduction to the Amazon EC2. The lab also guided us to create a VPC, new subnets, security groups, and launching an EC2 instance into the VPC.

Completion -

The screenshot shows a web browser window displaying the completion page for 'Lab 2: Build your VPC and Launch a Web Server' on the Vocareum platform. The page is titled 'Lab 2: Build your VPC and Launch a Web Server' and includes a section for 'Lab overview and objectives'. The objectives section states that the lab involves creating a VPC, adding subnets, creating a security group, and launching an EC2 instance to run a web server. The page also lists the tasks completed and their scores.

Task	Score
Total score	30/30
Task 1 - VPC created correctly	5/5
Task 2a - New subnets created correctly	5/5
Task 2b - Subnet route table association	5/5
Task 3 - Security group created correctly	5/5
Task 4a - EC2 instance created correctly	5/5
Task 4b - EC2 instance website accessible	5/5

LAB 3 Process -

Launching a Web Server Instance -

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-04d9d7a26a8b29d4b (Lab VPC)
10.0.0.0/16



Subnet [Info](#)

subnet-0cf5974f10f129f0a
VPC: vpc-04d9d7a26a8b29d4b Owner: 940016752965 Availability Zone: us-east-1a
Zone type: Availability Zone IP addresses available: 1 CIDR: 10.0.1.0/28

PublicSubnet1



[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply when outside of free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

Web Server security group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./!@,[@!+=&:[]\$*

Description - required [Info](#)

Security group for my web server

▼ Configure storage [Info](#)

[Advanced](#)

1x GiB Root volume 3000 IOPS (Not encrypted)

[i](#) Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage [×](#)

[Add new volume](#)

[?](#) Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.



0 x File systems

[Edit](#)

Termination protection [Info](#)

Enable

Stop protection [Info](#)

Select

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

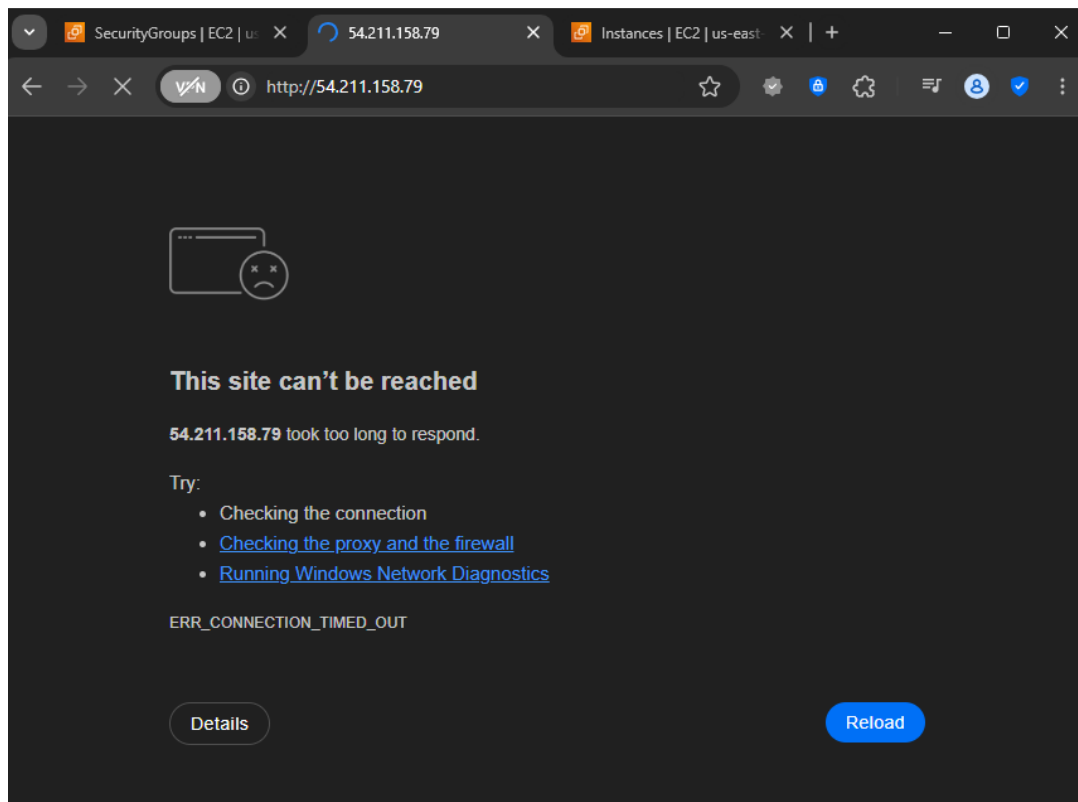
[Choose file](#)

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' > /var/www/html/index.htm|
```

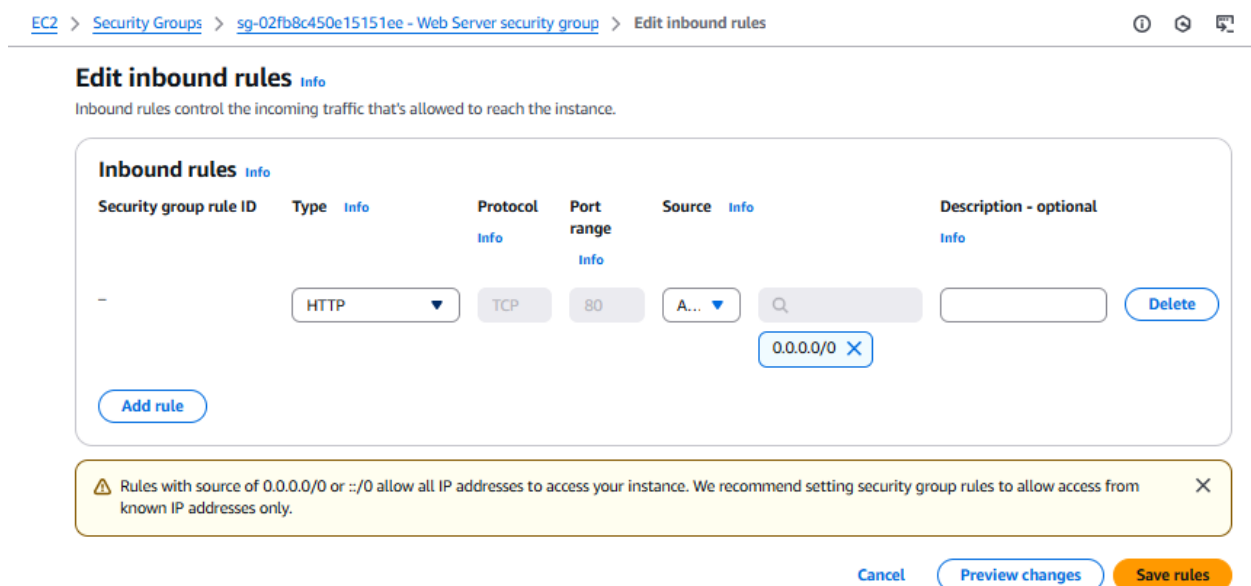
☐ User data has already been base64 encoded

Visiting the Web Server -

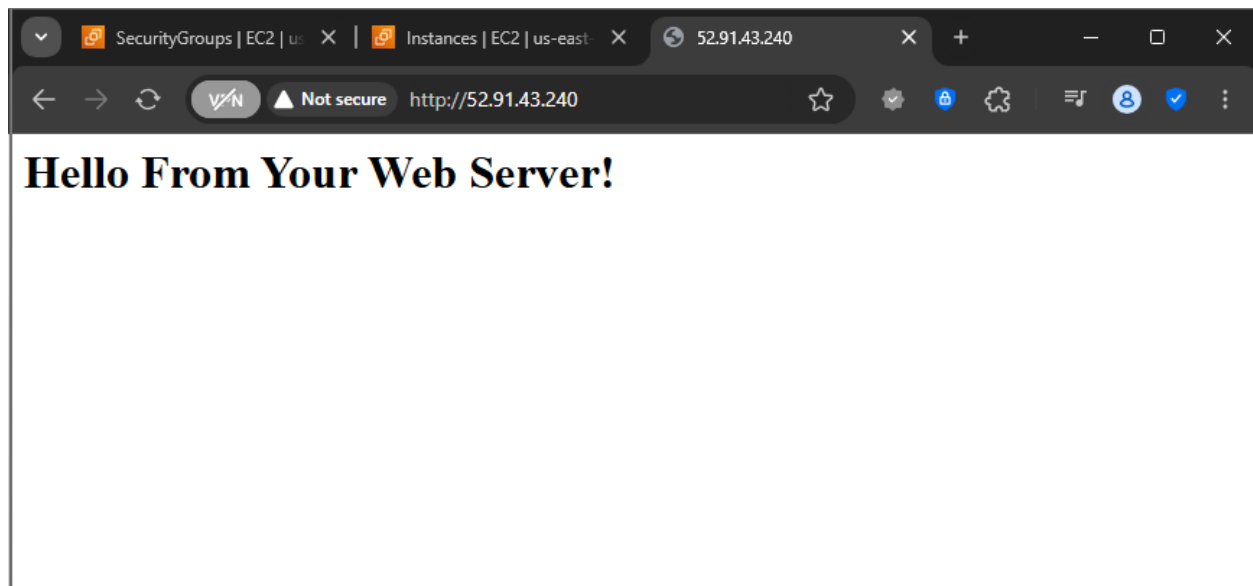
Initially, it will not work -



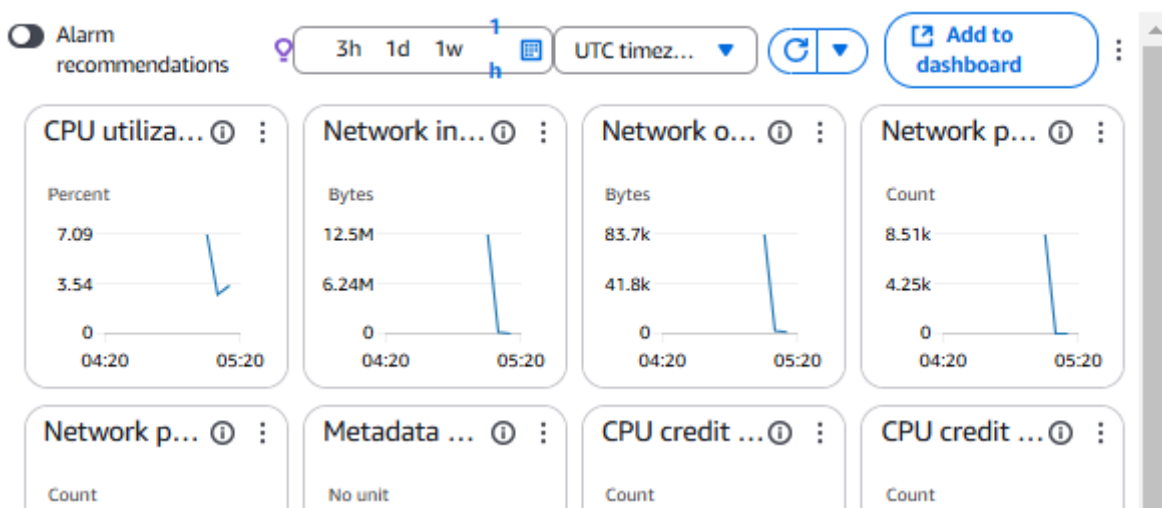
This is due to the security group blocking access to the Web server from port 80, http. Here we configure the security group to allow traffic from port 80 -



Now we are able to access the web server -



Monitoring the Web Server's Usages -



Modifying the size of the storage -

Modify volume [Info](#)

Modify the type, size, and performance of an EBS volume.

Volume details

Volume ID
[vol-030961a6e5e35ab51](#)

Volume type | [Info](#)
General Purpose SSD (gp3) ▼

Size (GiB) | [Info](#)

Min: 1 GiB, Max: 16384 GiB.

IOPS | [Info](#)

Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s) | [Info](#)

Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

[Cancel](#) [Modify](#)

Confirmation -

☒ Web Server i-03c6729d71d7a0302 Running t2.small 2/2 checks passed

i-03c6729d71d7a0302 (Web Server)

▼ Block devices

Filter block devices

☒	Volume ID	Device name	Volume size (GiB)	Attachment status	Attache
☒	vol-030961a6e5e35ab51	/dev/xvda	10	Attached	2025

Volume monitoring (1)

3h 1d 1w 1h UTC timezone

Add to dashboard

Average r... No unit

Average ... No unit

R... No unit

Write thro... No unit

Enabling Stop protection -

Info

Instance ID

 i-03c6729d71d7a0302 (Web Server)

Stop protection

☒ Enable

Cancel

Save

✕

Failed to stop the instance i-03c6729d71d7a0302
 The instance 'i-03c6729d71d7a0302' may not be stopped. Modify its 'disableApiStop' instance attribute and try again.

Notifications

 1 0 1 0 0

Instances (1/2) [Info](#)

Last updated less than a minute ago

All states

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check
☐	Bastion Host	i-0a87ecbdba3e60c7b	Running	t2.micro	2/2 checks passed
☒	Web Server	i-03c6729d71d7a0302	Running	t2.small	2/2 checks passed

i-03c6729d71d7a0302 (Web Server)

Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags

▼ Instance summary [Info](#)

Instance ID
 i-03c6729d71d7a0302

IPv6 address
—

Hostname type
IP name: ip-10-0-1-14.ec2.internal

Answer private resource DNS name

Public IPv4 address
 52.91.43.240 | [open address](#)

Instance state
 Running

Private IP DNS name (IPv4 only)
 ip-10-0-1-14.ec2.internal

Instance type

Private IPv4 addresses
 10.0.1.14

Public IPv4 DNS

ec2-52-91-43-240.compute-1.amazonaws.com
| [open address](#)

Elastic IP addresses

Created an EC2 instance and ran it as a web server, but initially it would not work because http traffic wasn't allowed. Creating a security group with inbound rules that allowed http fixed the issue and users are now able to access the web server.

Also experimented with monitoring the server's usages such as CPU usage and Metadata. Also experiment with different instance settings such as termination protection and stop protection.

Issues -

Multiple instances of error logging in, was basically locked out of AWS for the weekend, wasn't sure what to do until it fixed itself. Perhaps I was just simply timed out, or it may have been an AWS server-side issue.

Conclusion -

This lab introduces us to the Amazon Elastic Compute Cloud by allowed us to launch a web server with stop protection and termination protection, create and monitor an EC2 instance, modify and monitor security group to allow HTTP access, and to explore the limits of the Amazon EC2.

Completion -

The screenshot displays the Vocareum lab interface for 'Lab 3: Introduction to Amazon EC2'. The browser address bar shows the URL: <https://labs.vocareum.com/main/main.php?m=clabide&mode=s&asnid=3550738&stepid=3550739&hideNavBar=1>. The lab title is 'Lab 3: Introduction to Amazon EC2' with the subtitle 'Lab overview and objectives'. The main content area shows a diagram of the lab setup within an 'Availability Zone'. It includes a 'Security group' (red box) containing an 'IIS Web Server' (orange box) and an 'Amazon EC2' instance (orange box). The right sidebar shows the 'Total score' as 25/25 and a list of tasks with their completion status.

Task	Status
Total score	25/25
Task 1 - EC2 Instance created correctly	5/5
Task 2 - get system log requested	5/5
Task 3 - security group updated	5/5
Task 4 - EC2 Instance updated	5/5
Task 6 - Instance stopped on second try	5/5